

technology to volume manufacturing at low costs.

Acronym war in the mobile space

As of today, the OLED war is happening mainly in the mobile space. The air is filled with buzzwords ranging from Super-LCD and AMOLED to Super-AMOLED.

The addressing scheme used by OLED displays can be either passive-matrix (PMOLED) or active-matrix (AMOLED). The latter comprises a matrix of OLED pixels that generate light upon electrical activation. This is integrated with a thin-film transistor (TFT) array, which functions as a series of switches to control the current flowing to each individual pixel.

AMOLED displays consume less power than PMOLED displays, and are a step towards low-power, low-cost and large-size applications such as televisions. AMOLED technology is becoming quite popular in smartphones and digital cameras today. Certain manufacturers have started integrating the production of capacitive sensor arrays in the AMOLED module fabrication process, thereby adding touch capability to AMOLED displays. Samsung markets this technology as Super-AMOLED. Brightness and degradation issues have also been addressed to a certain extent in this new version.

AMOLED also lends itself to flexible displays. In fact, some AMOLED-based flexible display innovations were showcased at the *Consumer Electronics Show (CES) 2011*. One example is the 0.27mm thick, 11.43cm (4.5-inch) flexible WVGA AMOLED display with a bending radius of less than 10 mm showcased by Samsung Mobile Display. However, larger flexible displays are still on the wish-list.

The technology that is giving AMOLED stiff competition in the smart-phone space is Super-LCD (S-LCD). An upgrade of the time-tested LCD technology, this method of display is very power-efficient. It also overcomes some of the shortcomings of AMOLED, such

Hot, hotter, hottest

Mature and popular. Multi-touch and high-definition displays, large and small; LED and OLED-backlit LCD displays of all sizes; full OLED displays in mobile devices; low-power, eco-friendly displays

Mature and gaining popularity. MEMS-powered mobile displays; 3D display technology; heat-stabilised polyester, PET and PEN films and patterned transparent conductor (PTC) films for flexible displays

Being fine-tuned. Large-format OLED displays; interactive displays

Emerging. Surface-emissive display technologies; shapeless, transparent, flexible, low-power and interactive displays for ubiquitous computing

as the high proportion of white pixels on the screen. Some manufacturers appear to prefer S-LCD because it is based on an already mature technology and does not face degradation and other inherent problems of OLEDs. S-LCD is mainly promoted by Sony. Apple's Retina technology is also comparable to S-LCD.

Surface-emissive display technology

Surface-conduction electron-emitter display or surface-emissive display (SED) is a technology that was in development alongside LCDs. However, when LCDs gained dominance some years ago, many companies wound down their SED development efforts. When many thought SED was a dead technology, companies like AU Optronics and Prysm Display picked up the threads and gave life to it again. Now it is picking up steam, especially since it is believed that current developments in nanotechnology can breathe new life into SEDs.

SED technology is especially suited for flat-panel televisions and other thin displays. It uses nanoscopic electron emitters to energise coloured phosphors and produce an image. An SED can be seen as a matrix of tiny cathode ray tubes, each forming a single sub-pixel on the screen. These are grouped in threes, denoting red, green and blue (RGB) pixels. SEDs feature high contrast ratios, wide viewing angles, significantly low power consumption and very fast response times, all within a very thin form factor.

Prysm came out with a highly-discussed breakthrough in SEDs recently. Its new laser phosphor display (LPD) technology combines a phosphor

panel, laser engine and laser processor. The result is a thin, large-format display that combines the low-power consumption and reliability of solid-state lasers with a surface-emissive screen.

"Fundamentally different from LED or LCD-based display technology, the LPD creates a modular, highly configurable display panel with brilliant image quality and the industry's smallest environmental footprint. It uses rapid surface-scanning lasers controlled by advanced, high-speed inputs/outputs (I/Os) available on Xilinx field-programmable gate arrays (FPGAs) to excite phosphors, which, in turn, create brilliant, high-resolution pictures (with resolution as high as 4000 pixels) with very fast response," explains Raju.

In addition to brilliant picture quality, LPD brings the advantage of energy efficiency by using lasers for turning the pixels on or off compared to a conventional display with an always-on backlight.

Mastakar explains, "The innovations in LPD were made possible by developments in the FPGA technology. Advanced high-speed I/Os which can work in the range of 1.4 Gbps, innovative manufacturing processes that merge the advantages of high performance and low power, and integrated architectural blocks like memory controllers and multi-gigabit SERDES in Xilinx FPGAs make system design easier."

He adds, "LPD is one of the latest technologies in the SED category. It has the potential to replace backlit paper sign boards. It consumes less power, allows for true 180-degree viewing and provides incredible image quality.